Hepato-renal Function Assessment in Adult Male Wistar Rats Fed with White and Brown Species of *Phaseolus Vulgaris L* (Beans)

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ABSTRACT

This study assessed the hepato-renal functions following consumption of white or brown species of *Phaseolus vulgaris L* (beans) in rats. Twenty-five adults male Wistar rats (body weight 100 ± 20g) were used for this study. They were assigned to five groups as control fed normal rat chow (100g daily), groups 2 and 3 fed 25% and 50% white beans diet and groups 4 and 5 fed with 25% and 50% brown beans diet respectively for a 3 weeks duration. Blood samples were collected at day 22 and immediately centrifuged for estimation of serum Aspartate Aminotransferase (AST) Alanine Aminotransferase (ALT), Alkaline Phosphatase (ALP), protein, Blood Urea Nitrogen (BUN) and creatinine, using commercial kits. The liver and kidneys were harvested for relative weight determination. The results showed that dietary consumption of white and brown beans modulates hepato-renal indices in a variety- and dose-dependent manner. While both bean varieties showed a general trend toward hepatoprotection, evidenced by reductions in AST, ALT, and ALP, higher levels of brown beans (50%) led to an increase in liver and kidney weights and a rise in ALT activity, suggesting potential hepatic stress. Renal function was largely preserved across all groups, with significant reductions in serum creatinine in rats fed white beans. In conclusion, this study suggests that moderate consumption of white beans may support hepato-renal health, while excessive intake of brown beans should be approached with caution.

Keywords: *Phaseolus vulgaris*, Blood Urea Nitrogen, Creatinine, Liver enzymes.

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1. INTRODUCTION

The kidney and liver are vital organs responsible for maintaining homeostasis, metabolism, detoxification, and excretion of metabolic wastes. Hepato-renal health is important for overall physiological stability, and any impairment can lead to systemic disturbances and increased morbidity [1, 2]. The composition of diets play a large role in modulating liver and kidney function, as both organs are directly affected by the nature and quality of diet consumed.

Phaseolus vulgaris L., commonly known as the common beans, is one of the most widely consumed legumes globally, particularly in developing countries, due to its affordability and rich nutritional profile. It is a major source of protein, dietary fiber, vitamins, minerals, and bioactive compounds [3]. The two commonly consumed species; white beans and brown beans, have been reported to differ slightly in their phytochemical composition, which may influence their physiological and metabolic effects [4].

Despite their nutritional benefits, beans also contain antinutritional factors such as lectins, phytates, and tannins, which, if not adequately removed during processing, can pose risks to organ health [5]. These compounds may impair protein digestion and mineral absorption and, in large amounts, have been implicated in liver and kidney dysfunction [6]. Therefore, the long-term impact of consuming different

species of *P. vulgaris* on hepatic and renal function deserves investigation, particularly because of their widespread dietary use.

Following the increasing emphasis on plant-based diets and the major role of legumes like *Phaseolus vulgaris* in providing affordable nutrition, understanding their potential effects on major organs is essential. Studying the influence of white and brown bean diets on the hepato-renal system in Wistar rats will provide knowledge on their safety, potential toxicity, or protective effects, thus, informing dietary guidelines and health interventions, especially in populations that rely heavily on legumes for daily protein intake. While several studies have explored the nutritional value of beans, limited research has focused on their impacts on liver and kidney function, especially when comparing different species such as white and brown beans.

2. MATERIALS AND METHODS

2.1. Material: Two species of *Phaseolus vulgaris L* (white beans also referred to as Sokoto beans, and brown beans also known as honey beans), were purchased from Ekpoma local market, Edo State, Nigeria. Rat chow (Bendel feeds and flour meal Ewu, Nigeria) was obtained from a local store in Ekpoma. The materials and equipment used in this study include; plastic animal cages with top wire meshed, feeding troughs and drinkers, Electronic weighting scale, micropipette, plain tubes, sample and Ethylene Diamine Tetra Acetic acid (EDTA) bottles, synthetic hand gloves, wooden dissecting board, dissecting sets, syringes and needles, cotton wool, filter paper, measuring cylinder, beakers, disinfectants and detergents (for washing and cleaning), ice block, cooler, wooden shadings for animal's beddings. Also included are normal saline, formalin and other chemicals were of analytical grade.

2.2. Care and management of Wistar rats: Twenty-five adults male Wistar rats of comparable weights (100 ± 20g) were used for this study. The rats were obtained from the Animal House of the College of Medicine, Ambrose Alli University, Ekpoma and transferred to the Department of Physiology animal holding facility in the same institution. They were housed in plastic cage with a wire screen top. The animals were fed *ad libitum* with clean tap water and pelleted rat chow during acclimatisation (7 days) and were maintained under standard care of laboratory animals as stated in the guidelines and regulations for the use of animals in scientific experiment.

2.3. *Phaseolus vulgaris* preparation: A bowl each of white and brown beans were washed with clean water separately. Thereafter, they were put in iron pot and heated for 50 to 60 minutes adding adequate amount of water (3.5 to 5.5 litres) to cook the beans. After cooking, they were allowed to cool and then air dried. After drying they were grounded using electronic machine into powder and were used for the diets formulations.

2.4. Diet formulations: The preparation of the 25% and 50% of the white and brown bean diets was as described by Nwaopara *et al*., [7]. Briefly, to prepare 3000g of the 25% white or brown beans diet, 750g of the grinded white or brown beans was mixed with 2250g of normal rat chow and for the 50% white or brown beans diet, 1500g of the grounded white or brown beans was mixed with 1500g of normal rat chow. These were thoroughly mixed by adding adequate clean water to form paste. The pastes were cut into small pieces (approximately 2-4mm diameter) and allowed to dry under the sun. The sun-dried mixtures were then packed into clean sack bag for use.

2.5. Experimental grouping: After acclimatization, the rats were assigned to five (5) groups of five rats each as follows.

Group 1 (control group): Comprised of 5 male rats fed normal rat chow (100g daily) and clean water only throughout the study.

Group 2: Comprised of 5 male rats fed formulated 25% white beans diet (100g daily) and clean water alone throughout the study.

Group 3: Comprised of 5 male rats fed formulated 50% white beans diet (100g daily) and clean water alone throughout the study.

Group 4: Comprised of 5 male rats fed formulated 25% brown beans diet (100g daily) and clean water alone throughout the study.

Group 5: Comprised of 5 male rats fed formulated 50% brown beans diet (100g daily) and clean water alone throughout the study.

2.6. Feeding duration: Rats in all the groups received *ad libitum* the white and brown beans diets based on corresponding diet group. They were given the different diets daily between the hours of 8:00 am and 10:00 am. Next day leftover diets and water were discarded while rats bedding was changed every other day. The administration of the different beans' diets lasted for 21 days.

2.7. Sample collection: At the end of the treatment (22nd day of diet administration), the rats were cervically decapitated and blood sample collected following standard laboratory procedures. The rats were dissected along the lateral side of the neck to assess the jugular blood vessels. Blood sample was then obtained using 2mL syringe and the blood collected into EDTA bottles. The kidneys and liver were also harvested and weighed using electronic weighting balance and the relative weight calculated.

2.8. Sample analysis: The blood sample obtained was centrifuged for serum concentration of BUN, creatinine, AST, ALT, and ALP using commercial kits (Randox Laboratories kits, United Kingdom). These parameters were estimated from serum samples as per the procedure described in the kit.

2.9. Data analysis: The mean ± SEM were determined and ANOVA statistical test performed using SPSS version 20 software. The value $p<0.05$ was considered statistically significant.

3. RESULTS

Relative weight of the liver and kidneys in rats fed white and brown *Phaseolus vulgaris*.

Figure 1 is a comparative chart of the relative weight of the liver and kidneys following white and brown *Phaseolus vulgaris* ingestion. White or brown beans was observed to increase the relative weights of the liver and kidneys with significant different in the 50% brown beans compared with the control diet.

Serum total protein and liver enzymes activities in fed white and brown *Phaseolus vulgaris.*

Figure 2 presents a comparative chart of serum total protein and liver enzymes in adult Wistar rats fed white and brown species of *Phaseolus vulgaris L*. Ingestion of white or brown beans did not significantly alter the serum total protein compared with the control diet. However, the ingestion of white or brown beans resulted in a decrease in AST, ALT and ALP activities with significant different in the 25% and 50% white beans and 25% brown beans compared with the control diet.

Blood urea nitrogen and serum creatinine level in rats fed white and brown species of *Phaseolus vulgaris*

Figure 3 represents blood urea nitrogen and serum creatinine levels in adult Wistar rats fed white and brown species of *Phaseolus vulgaris*. There was no significant different ($p>0.05$) in the blood urea nitrogen level following the ingestion of white or brown beans compared with the control diet. White or brown bean ingestion resulted in decrease in serum creatinine level. However, only the 25% and 50% white beans groups presented significant decrease ($p<0.05$) in serum creatinine level compared with the control diet.

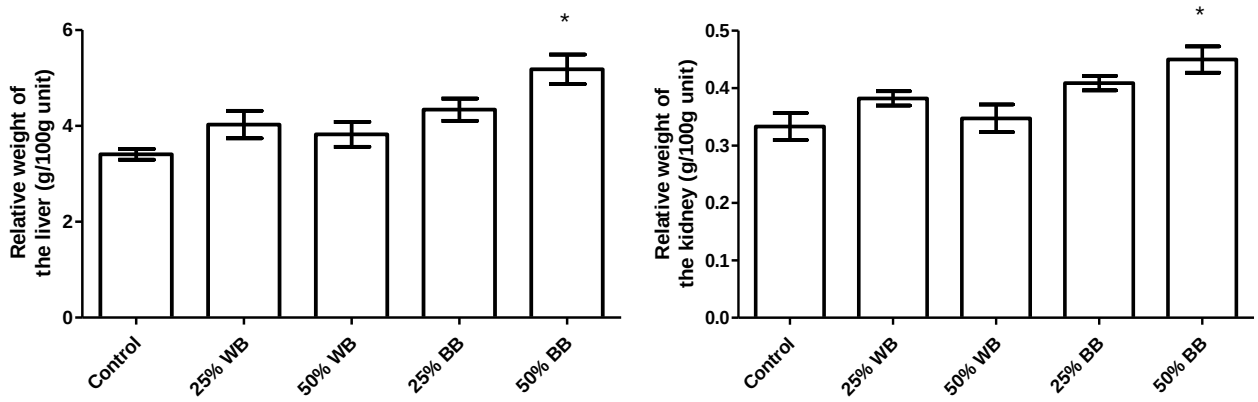


Figure 1: The relative weight of the liver and kidney in adult Wistar rats fed white and brown species of *Phaseolus vulgaris*.

Values are means ± SEM, n = 5, * statistically significant compared with control diet. Key: 25% WB = 25% white beans, 50% WB = 50 % white beans, 25% BB = 25% brown beans, 50% BB = 50% brown beans.

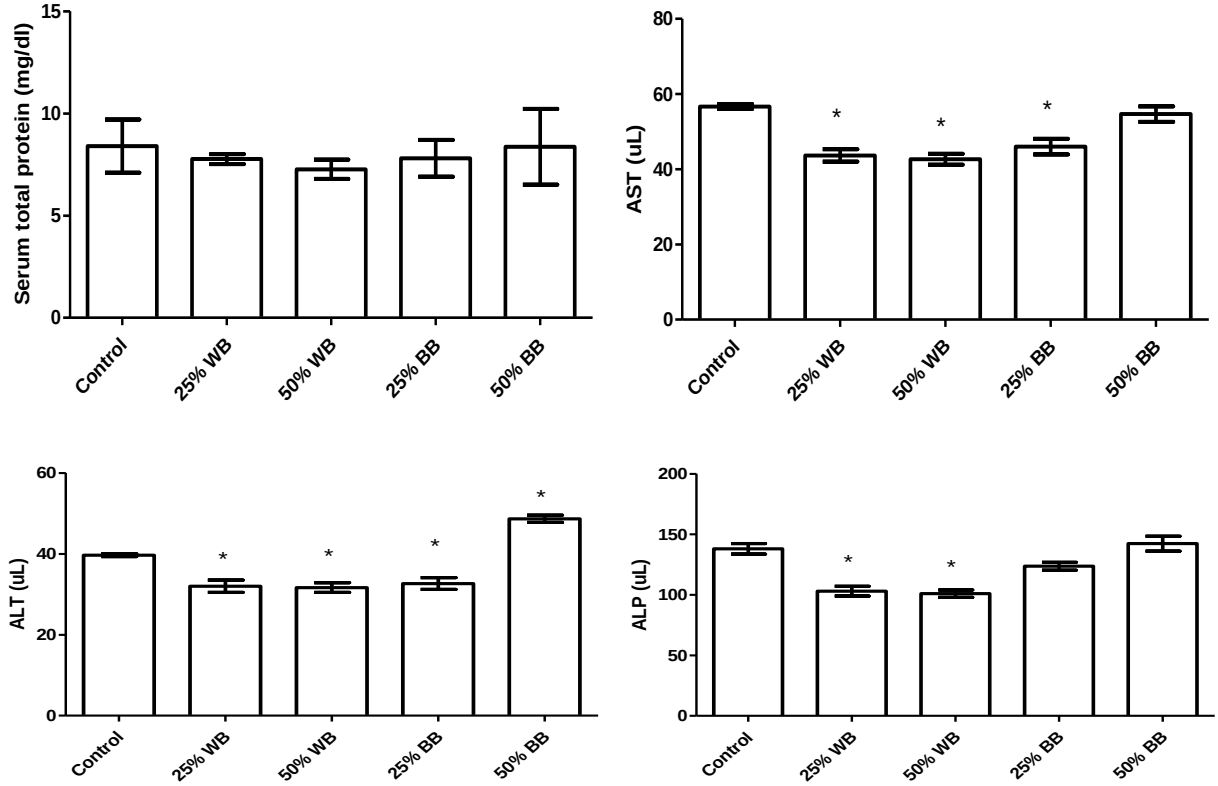


Figure 2: Serum total protein level and liver enzymes activities in rats fed white and brown species of *Phaseolus vulgaris*.

Values are means ± SEM, n = 5, *statistically significant compared with control diet. Key: 25% WB = 25% white beans, 50% WB = 50 % white beans, 25% BB = 25% brown beans, 50% BB = 50% brown beans

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4. DISCUSSION

This study investigated the impact of dietary consumption of white and brown beans on hepatic and renal functional profiles in adult male Wistar rats. The observed differences in organ weights and biochemical markers offer important information on the physiological responses to beans consumption, with significant distinctions between the white and brown varieties, as well as the graded consumption levels. The significant increase in relative liver and kidney weights, particularly in the 50% brown bean group, suggests a potential hypertrophic response possibly due to heightened metabolic or detoxification demands. The liver, which is the primary site for metabolism and protein synthesis, may have responded to phytochemicals or antinutritional factors in the beans. This response was more pronounced with brown beans, which are known to contain higher levels of certain bioactive compounds, including polyphenols and tannins [8]. Similarly, the increased kidney weight might reflect a compensatory response to elevated solute load or subtle nephrotoxicity, although the functional indices remained largely unaltered.

The decrease in serum AST and ALT activities following consumption of white and brown beans (25% and 50%) indicates a potential hepatoprotective impact, especially with white beans. The exception was the significant increase in ALT activity in the 50% brown bean group, which could suggest hepatocellular stress or mild hepatic injury at higher doses of brown beans. ALT is more liver-specific than AST and its elevation may point to selective hepatocellular membrane leakage or mitochondrial dysfunction [9]. The differential effect on ALP activity, reduced by white beans but unaffected by brown beans, further supports the notion of variety-dependent hepatic impact, possibly linked to the bioavailability of compounds affecting biliary function or enzyme synthesis.

There were no significant alterations in serum total protein, suggesting that protein synthetic function of the liver was largely maintained despite changes in enzyme (transaminase) activity. This indicates preserved hepatic biosynthetic capacity and aligns with the observed decrease in liver enzymes in most groups, reflecting an absence of overt hepatic dysfunction.

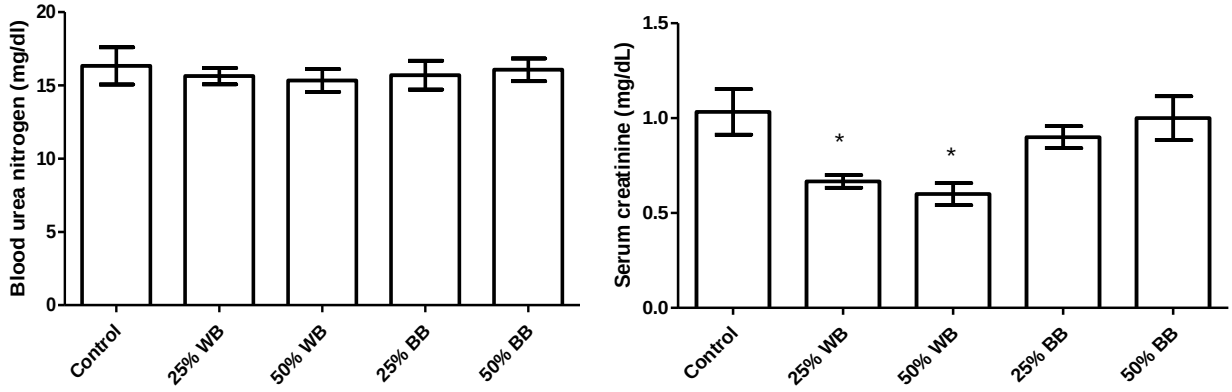


Figure 3: Blood urea nitrogen and serum creatinine level in rats fed white and brown species of *Phaseolus vulgaris*.

Values are means ± SEM, n = 5, *statistically significant compared with control diet. Key: 25% WB = 25% white beans, 50% WB = 50 % white beans, 25% BB = 25% brown beans, 50% BB = 50% brown beans.

The lack of significant changes in blood urea nitrogen and the reduction in serum creatinine levels, particularly with 25% and 50% white beans is suggestive of the fact that consumption did not adversely affect renal function. Instead, the reduced creatinine may indicate enhanced renal clearance or decreased muscle catabolism, both of which can occur with dietary modifications. The absence of significant electrolyte disturbances (Na^+, Cl^-) further confirms preserved renal electrolyte handling. These findings are in agreement with previous reports on the beneficial effects of *Phaseolus vulgaris* on metabolic parameters and organ protection, including anti-inflammatory and antioxidant properties [10, 11].

5. CONCLUSION

The findings of this study suggests that moderate consumption of white and brown beans may support hepato-renal health, while excessive intake of brown beans should be approached with caution. The increased liver and kidney weights, coupled with the ALT rise in the 50% brown bean group, warrant caution, particularly with high levels of brown beans intake.

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